

The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL 1 MODULE, AUTUMN SEMESTER 2016-2017

MATHEMATICS FOR COMPUTER SCIENTISTS

Time allowed ONE hour and THIRTY minutes only

Candidates must NOT start writing their answers until told to do so

There are THREE questions in the exam paper. Answer all the THREE questions.

Only silent, self-contained calculators with a single-line display are permitted in this examination.

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

Model Solutions

K – Knowledge
C – Comprehension
P – Problem solving

1. Questions on Number theory, Logic, Sets, and Functions.

(a) For a non-empty tuple of positive integers $a_1, a_2, \dots, a_n$, let

$$\text{CD}(a_1, a_2, \dots, a_n) = \{d \in \mathbb{N} : \forall 1 \leq i \leq n, d|a_i\}$$

be the set of natural numbers that are common divisors of all $a_1, a_2, \dots, a_n$.(i) Using the Fundamental Theorem of Arithmetic (FTA), find $\text{CD}(420, 200, 315)$.

[4 marks]

Answers: (K)

$$420 = 2^2 * 3^1 * 5^1 * 7^1; [1 \text{ mark}]$$

$$200 = 2^3 * 3^0 * 5^2 * 7^0; [1 \text{ mark}]$$

$$315 = 2^0 * 3^2 * 5^1 * 7^1; [1 \text{ mark}]$$

This amounts to finding for each prime number its minimum multiplicity. Accordingly, we have $2^0 = 3^0 = 7^0 = 1$, and $5^1 = 5$. Thus $\text{CD}(420, 200, 315) = \{1, 5\}$ [1 mark]

[If no evidence of using the FTA but the final answer is correct, then award 2 marks]

(ii) Prove or give a counter example that for positive integers a_1, a_2, a_3 , if $\text{CD}(a_1, a_2, a_3) = \{1\}$, then $\text{CD}(a_1 \cdot a_2, a_3) = \{1\}$, where $a_1 \cdot a_2$ denotes the multiplication of a_1 and a_2 .

[6 marks]

Answers: (C)

We give a counter example. Assume $a_1 = 1$, $a_2 = 2 \cdot b$, $a_3 = 3 \cdot b$, where $b > 1$ is a positive integer. Clearly, $\text{CD}(a_1, a_2, a_3) = \{1\}$. However, since $a_1 \cdot a_2 = a_2 = 2 \cdot b$, thus b divides $a_1 \cdot a_2$. Also b divides a_3 . Therefore $\text{CD}(a_1 \cdot a_2, a_3)$ contains at least 1 and b , i.e., $\{1, b\}$ is a subset of $\text{CD}(a_1 \cdot a_2, a_3)$.

23 marks for recognising that there exist counter examples. **23** marks for any correct construction of counter examples. **2** marks for explanation.

(iii) Without using the FTA, prove that for positive integers a and a' , if $\text{CD}(a, a') = \{1\}$ then, for all integers k ,

$$(a \cdot a')|k \leftrightarrow a|k \wedge a'|k$$

[7 marks]

Answers: (C, P)

Firstly, let's prove that $(a \cdot a')|k \rightarrow a|k$ and $a'|k$. From the antecedence, we know that $k = a \cdot a' \cdot n$ for some integer n . [1 mark]

Thus $k = a*(a'*n) = a'*(a*n)$. Hence the consequence. [1 marks]

Conversely, we prove that $a|k$ and $a'|k \rightarrow (a*a')|k$. From the antecedence, $k = a*n = a'*m$, for some integers n and m . [1 marks]

Since a and a' are relatively prime (i.e., having the common divisor 1), then integer n must be a multiple of a' . [1 mark]

Similarly, m must be a multiple of a . [1 mark]

This leads to $k = a*(a'*x) = a'*(a*y)$. [1 mark]

It follows that $x=y$ and $(a*a')$ divides k . [1 mark]

Any other alternative methods that are provided with steps are also acceptable and will be awarded marks accordingly.

(b) Let $X_n = \{p_1 = 2, p_2 = 3, p_3 = 5, \dots, p_n\}$ be the set of the first n prime numbers. For example, $X_5 = \{2, 3, 5, 7, 11\}$. Let $S_n^{(m)} = \{a \in \mathbb{N} : \forall 0 \leq i \leq n, \exists 0 \leq r_i \leq m \in \mathbb{N}, p_i \in X_n, a = p_1^{r_1} \cdot p_2^{r_2} \cdot \dots \cdot p_i^{r_i} \cdot \dots \cdot p_n^{r_n}\}$ be the set of integers that can be factorised into the multiplication of the first n prime numbers, where the multiplicity of each prime number is less than or equal to a positive integer m .

(i) Given an example of ~~the-an~~ integer $a \in S_5^3$ and explain why it is valid.

[4 marks]

Answers: (k)

$$a = 2^1 * 3^0 * 5^3 * 7^0 * 11^0 = 10250.$$

[2.5 marks for correct use of prime numbers 2, 3, 5, 7, 11]

[1.5 marks for correct use of the multiplicity 0, 1, 2, 3]

(ii) Specify the cardinality of $S_n^{(m)}$. Show your work.

[4 marks]

Answers: (C)

For each prime number, the multiplicity varies from 0 to m , thus in total $(m+1)$ choices.

[2 marks]

So there are in total $n^{(m+1)}$ different numbers in $S_n^{(m)}$. [2 marks]

(iii) Let $\mathcal{P}(X_n)$ denote the power set of X_n . Define the function $f: S_n^{(m)} \rightarrow \mathcal{P}(X_n)$ such that

$$f(a) = \{p_i \in X_n : \forall 0 \leq i \leq n, r_i > 0, a = p_1^{r_1} \cdot p_2^{r_2} \cdot \dots \cdot p_n^{r_n} \in S_n^{(m)}\}.$$

Show that f is surjective (i.e., onto) but not injective (i.e., not one-to-one).

[9 marks]

Answers: (C)

Firstly, we show that f is surjective. That is to prove that for any subset Y of $\{p_1, p_2, \dots, p_n\}$, there exist at least one number a in $S_n^{(m)}$, such that $f(a) = Y$. [± 1.5 mark]

Given Y , we can construct a in such a way that it is the multiplication of all the elements in Y . Thus the multiplicity of each prime number is either 0 or 1, i.e., between 0 and m . [2 marks, any other valid construction is also acceptable]

According to the definition of $S_n^{(m)}$, such construction of a is one of its members and also satisfy $f(a)=Y$. [1 mark]

Secondly, we show that f is not injective. That is to disprove that if a is not equal to b , then $f(a)$ is not equal to $f(b)$. [1 mark]

We can give a counter example, i.e., show that for a is not equal to b , we have $f(a)=f(b)$. [± 1.5 mark]

Construct a and b such that they share the same prime numbers as common divisors but only differ in their multiplicities. [1 mark]

According to the definition of f , we have $f(a) = f(b)$. [1 mark]

~~[1 mark for the clarity]~~ Any other alternative proofs that provide reasonable steps are also acceptable and will be awarded marks accordingly.

2. Questions on Logic, Sets, Induction, and Relations.

(a) A set S is defined recursively by

Basic step: $0 \in S$

Recursive step: if $a \in S$, then $a + 3 \in S$ and $a + 5 \in S$.

(i) Determine the set $S \cap \{a \in \mathbb{Z}: 0 < a < 12\}$.

[5 marks]

Answers: (k)

List the elements in $\{a \in \mathbb{Z}: 0 < a < 12\}$ until 12. [1 mark for the strategy and alike]

0

$0+3 = 3, 0+5=5$ [1 mark]

$3+3=6, 3+5=8, 5+3=8, 5+5=10$ [1 mark]

$6+3=9, 6+5=11, 8+3=11, 8+5=13, \dots$ [1 mark]

Thus the intersection is $\{3,5,6,8,9,10,11\}$ [1 mark]

(ii) Prove that every integer $a \geq 8$ is contained in S . (Hint:)

[8 marks]

Answers: (C)

Use mathematical induction. Recognise that we are to prove the proposition for all $a \geq 8$, $P(a)$ is true where $P(a)$ represents that "a is contained in S ". [1 mark]

Basic step: prove $P(8), P(9), P(10), P(11)$. [1 mark]

I.e., prove that $8 \leq a \leq 11$ are contained in S . This is trivial from the results in (i). [1 mark]

Inductive step:

Assume $P(n), P(n-1), P(n-2), P(n-3)$ for $n \geq 11$ are true. [1 mark]

That is to assume that $n-3, n-2, n-1$ are contained in S . [1 mark]

Prove that $P(n+1)$ is true. That is to prove that $n+1$ is contained in S . [1 mark]

According to the assumption we know that $n-2$ is contained in S , [1 mark]

thus according to the recursive definition of S , $n-2+3=n+1$ is also contained in S . [1 mark]

(b) Let $X = \{x, y\}$. Define a binary relation σ on $\mathcal{P}(X)$ the power set of X , by $A \sigma B$ if and only if $A \subseteq B$.

(i) Specify $X \times X$ and $|\mathcal{P}(X)|$.

[4 marks]

Answers: (k)

$X \times X = \{(x,x), (x,y), (y,x), (y,y)\}$ [2 marks]

The cardinality of the power set of X is $2^2 = 4$. (i.e., each element of X is either included or not included in a set. [2 marks]

(ii) How many different binary relations can be defined on the set X ? Show your work.

[4 marks]

Answers: (C)

A binary relation on the set X is a subset of the Cartesian product $X \times X$. [1 mark]

The number of different binary relations is the same as the cardinality of the power set of $X \times X$. [1 mark]

According to (i), there are four elements in $X \times X$, thus the cardinality of its power set is $2^4 = 16$. [2 marks]

(iii) Is σ reflexive, antisymmetric, and/or transitive? Explain your answer.

[6 marks]

Answers: (K, C)

It is reflexive [1 mark], since A is a subset of A [1 mark]

It is antisymmetric [1 mark], since if A is a proper subset of B , then B can't be a proper subset of A [1 mark]

It is transitive [1 mark], since if A is a subset of B , and B is a subset of C , then A must be a subset of C [1 mark]

(iv) Represent the relation σ as a Boolean matrix $\mathbf{M}_\sigma$, and determine $\mathbf{M}_\sigma^2$. Show your work.

[6 marks]

Answers: (K)

	Null	{x}	{y}	{x,y}
Null	1	1	1	1
{x}	0	1	0	1
{y}	0	0	1	1
{x,y}	0	0	0	1

[4 marks]

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}^2 \\
 = \\
 \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}$$

[0 1 0 1]

[0 0 1 1]

[0 0 0 1]

[2 marks]

3. Questions on Sets, Counting, and Probability.

(a) Suppose that A and B are events from a sample space S . Denote the probability of A and B as $p(A)$ and $p(B)$ respectively.

- (i) Let $S = \{s_1, s_2, s_3, s_4, s_5\}$, $A = \{s_1, s_2, s_4\}$, and $B = \{s_2, s_5\}$. Assuming each sample $s_i, i = 1, \dots, 4$ is equally likely but s_5 happens as twice likely as the rest of the samples. Determine the values of $p(A)$ and $p(B)$ respectively. Justify your answer.

[5 marks]

Answers: (K)

$$P(s_1)=p(s_2)=p(s_3)=p(s_4)=1/2 \cdot p(s_5) \quad [1 \text{ mark}]$$

$$P(s_1)+p(s_2)+p(s_3)+p(s_4)+p(s_5)=1 \quad [1 \text{ mark}]$$

$$\text{Thus } P(s_1)=p(s_2)=p(s_3)=p(s_4)=1/6, \quad p(s_5)=2/6=1/3. \quad [1 \text{ mark}]$$

$$\text{Thus } p(A) = p(s_1)+p(s_2)+p(s_4)=1/2 \quad [1 \text{ mark}]$$

$$P(B) = p(s_2) + p(s_5) = 1/2 \quad [1 \text{ mark}]$$

- (ii) Under the same conditions given in (i), determine the conditional probability $p(A|B)$. Justify your answer.

[4 marks]

Answers: (K)

$$A \text{ intersection } B = \{s_2\} \quad [2 \text{ mark}]$$

$$P(A|B) = P(A,B)/p(B) \quad [1 \text{ mark}]$$

$$= p(s_2)/p(B) = 1/3 \quad [1 \text{ mark}]$$

- (iii) Given that $p(A) \neq 0$ and $p(B) \neq 0$, show that if $p(B|A) < p(B)$, then $p(A|B) < p(A)$.

[7 marks]

Answers: (C)

$$P(A,B) = p(A|B) \cdot p(B) \quad [2 \text{ mark}]$$

$$= p(B|A) \cdot p(A) \quad [2 \text{ mark}]$$

$$\text{Thus we have } p(B|A)/p(B) = p(A|B) \cdot p(B) \quad [1 \text{ mark}]$$

$$\text{Since } p(B|A) < p(B), \text{ then } p(B|A)/p(B) < 1, \quad [1 \text{ mark}]$$

$$\text{Accordingly, } p(A|B) \cdot p(B) < 1, \text{ and thus } p(A|B) < p(B) \quad [1 \text{ mark}]$$

(b) A committee of five is to be chosen randomly from a collection of 4 women and 8 men.

(i) How many different committees of five people can be formed? Justify your answer.

[5 marks]

Answers: (C)

$4+8 = 12$ people to be chosen from in total. [2 marks]

$C(12,5)$ [2 marks for the correct formula and 1 mark for the final answer]

(ii) How many different committees of five can be formed, if at least two men and at least two women are to be on the committee of five? Justify your answer.

[8 marks]

Answers: (C)

Only two cases.

2 men + 3 women. [0.5 mark]

This gives $C(8,2)*C(4,3)$ [3 marks for the correct formula] - (1)

Or 3 men + 2 women. [0.5 mark]

This gives $C(8,3)*C(4,2)$ [3 marks for the correct formula] - (2)

(1) +(2) gives the final answer. [1 mark for the final answer] - (3)

(iii) What is the probability that there are exactly 3 women and two men chosen, given that the constraints in part (ii) hold?

[4 marks]

Answers: (C)

Recognise that this is to find the number of choices for 3 women + 2 men over the total number of choices as the result in (ii). [2 marks]

$(1)/(3)$ as in (ii). [2 marks]